

Requirement Gathering

GITHUB LINK : <https://github.com/daniellajaber-cs/IT-HelpDesk-Ticketing-System/tree/main>

PROJECT OVERVIEW

The IT Help Desk & Ticketing Management System is a web-based platform designed to help companies/organizations to manage requests (technical) effectively.

Employees can submit their tickets, while an IT agent and administrators can resolve the issues they have, track, assign them with solutions and manage the issues through an organized dashboard . The system improves communication , organization and support workflow management inside the organisation.

PROBLEM STATEMENT

Many companies still handle technical issues manually like through emails, messaging applications, which is not effective and can lead to lost requests, delays ,poor tracking and miscommunication .

The proposed system provides an organized solution that supports operations ,improves ticket tracking and technical issues effectively .

SYSTEM OBJECTIVES

These are the goal of the system :

- Build an organized ticket management system
- Improve communication between employees and IT support
- Organize the workflow system
- Track assigned ticket status
- Generate reports and analytics on each issue
- Improve response and resolution time
- Provide role-based system access
- Integrate AI-powered (AI Agent) for automation features

USER & ROLES

Admins

- Full system access
- Manages users and roles
- Assign tickets to employees
- Generates report on each issue
- Manages categories/settings

IT Support Agent

- Views assigned tickets from employees

- Updates ticket Status
- Resolves technical issues
- Add comments and notes regarding the issues

Employee

- Create support tickets
- Tracks ticket Progress
- Uploads attachments regarding their issue
- Receives notifications

Manager

- Monitors reports and analytics
- Reviews team performance
- Tracks support activities

FUNCTIONAL REQUIREMENTS

Authentication Requirement

- User login
- Password encryption
- Forget /reset password
- Role access control
- Session management
- Protected API routes

Ticket Management Requirement

- Create support tickets
- Update tickets
- Delete tickets
- Track ticket history
- Search and filter tickets
- Generate ticket reference number
- Manage categories, priorities and statuses

Assignment and Workflow Requirements

- Assign tickets to agents
- Assign tickets
- Internal comments and notes
- Track assignment history

Notification Requirements

- In-app notification
- Email notification

- Ticket update alerts
- Comment and reply system

Dashboard and reporting Requirements

- View ticket statistics
- Generate monthly reports
- Display charts and analytics
- Export report to PDF and Excel

File Attachment Requirements

- Upload screenshots or documents
- Secure file storage
- File type validation
- File size validation

Admin Panel Requirements

- Manage users and roles
- Manage ticket categories
- Monitor system activity
- Configure system settings

AI Features Requirements

- Categorized AI tickets
- Suggested AI priority
- AI troubleshooting suggestions
- AI chat assistant

NON-FUNCTIONAL REQUIREMENTS

- Responsive and mobile friendly design
- Fast system performance
- Secure authentication and authorization
- Scalable architecture
- Organized and user friendly interface
- Reliable database management
- Error handling and validation
- Maintainable and modular code structure

SYSTEM MODULES

Module	Description
Authentication Module	Handles login, registration, and security
Ticket Management Module	Handles ticket creation and updates
Workflow Module	Handles assignment and escalation
Notification Module	Sends alerts and updates
Dashboard Module	Displays reports and analytics
Admin Module	Controls users and settings
AI Module	Provides smart automation features

TECHNOLOGIES USED

Layer	Chosen Technology
Frontend	React.js (JavaScript)
Backend	ASP.NET Core Web API (C#)
Database	SQL Server +SSMS
Version Control	Git &Github
API Testing	Postman
UI/UX Design	Google Stitch
Database Design	Draw.io

EXPECTED OUTPUTS

- Modern IT Help Desk web application
- Secure ticket management system
- Admin and employee dashboards
- Reporting and analytics system
- Organized database structure
- Responsive UI/UX
- AI-assisted support feature